Project 3:-Maven Selenium Project – Automating Web Applications Using Chrome Browser

Objective

To create a **Maven-based Selenium project using the Firefox browser** that performs the following tasks:

- Login to the SauceDemo website
- Search and add a product to the cart on AutomationExercise website
- Login to the Practice Test Automation website
- Demonstrate **multi-tab browser automation using Selenium WebDriver**

Websites Used for Automation

The following websites are used in this project:

1. SauceDemo
   https://www.saucedemo.com/
2. Automation Exercise
   https://automationexercise.com/products
3. Practice Test Automation
   https://practicetestautomation.com/practice-test-login/

Step 1: Create a Maven Project

Open the terminal and create a Maven project using the following command:

mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumApp01 -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false

NOTE: Similary create maven project for website 2 and 3
MyMavenSeleniumApp02
MyMavenSeleniumApp03

```
brijesh@Ubuntu:~/devops$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumApp01 -Darchetype
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/brijesh/devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenSeleniumApp01
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/brijesh/devops/MyMavenSeleniumApp01
[INFO] ----------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ----------------------------------------------------------------------------
[INFO] Total time:  3.067 s
[INFO] Finished at: 2026-03-21T14:18:43+05:30
[INFO] ----------------------------------------------------------------------------
brijesh@Ubuntu:~/devops$
```

Step 2: Install ChromeDriver

Before running Selenium scripts, ChromeDriver must be installed.

Check Installed Chrome Version

Run the following command:

google-chrome --version

Example Output:

Google Chrome 133.0.6943.141

Extract Chrome Version

CHROME_VERSION=$(google-chrome --version | awk '{print $3}' | cut -d'.' -f1-3)

This extracts the Chrome version and stores it in the variable **CHROME_VERSION**.

Install jq (Used for JSON Parsing)

sudo apt install jq

Install curl (If Not Installed)

sudo apt install curl

Get Latest Compatible ChromeDriver Version

LATEST_DRIVER=$(curl -s "https://googlechromelabs.github.io/chrome-for-testing/latest-patch-versions-per-build.json" | jq -r ".builds[\"$CHROME_VERSION\"].version")

Download ChromeDriver

wget https://storage.googleapis.com/chrome-for-testing-public/$LATEST_DRIVER/linux64/chromedriver-linux64.zip

Extract ChromeDriver

unzip chromedriver-linux64.zip

Move ChromeDriver to System Path

sudo mv chromedriver-linux64/chromedriver /usr/local/bin/chromedriver
sudo chmod +x /usr/local/bin/chromedriver

Verify ChromeDriver Installation

chromedriver --version

Example Output:

ChromeDriver 133.0.6943.141

The version should match your Chrome version.

Step 3: Add Selenium Dependency in pom.xml

Open the **pom.xml** file and add the Selenium dependency.

```
<dependency>
<groupId>org.seleniumhq.selenium</groupId>
<artifactId>selenium-java</artifactId>
<version>4.29.0</version>
</dependency>
```

<build>

```
<plugins>
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.7</source>
<target>1.7</target>
</configuration>
</plugin>
<!-- Example: Maven Surefire Plugin to run tests -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-surefire-plugin</artifactId>
<version>2.22.2</version>
</plugin>
<plugin>
        <groupId>org.apache.maven.plugins</groupId>

        <artifactId>maven-jar-plugin</artifactId>
        <version>3.0.1</version>
       <configuration>
           <archive>
               <manifestEntries>
                   <Main-Class>com.example.App</Main-Class></manifestEntries>
</archive>
 </configuration>
</plugin>
</plugins>
</build>
</project>
```

pom.xml
~/Devops/MyMavenSeleniumApp02

```xml
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.example</groupId>
  <artifactId>MyMavenSeleniumApp01</artifactId>
  <packaging>jar</packaging>
  <version>1.0-SNAPSHOT</version>
  <name>MyMavenSeleniumApp01</name>
  <url>http://maven.apache.org</url>


  <dependencies>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>3.8.1</version>
      <scope>test</scope>
    </dependency>


    <!-- https://mvnrepository.com/artifact/org.seleniumhq.selenium/selenium-java -->
<dependency>
    <groupId>org.seleniumhq.selenium</groupId>
    <artifactId>selenium-java</artifactId>
    <version>4.29.0</version>
</dependency>
  </dependencies>



  <!-- Build: Configuring plugins and build settings -->
<build>
<plugins>
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.7</source>
<target>1.7</target>
</configuration>
</plugin>
<!-- Example: Maven Surefire Plugin to run tests -->
<plugin>
```

Step 4: Create App.java

Location:

src/main/java/com/example/App.java

1.App.java code for AutomationExercise

```
package com.example;

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.interactions.Actions;

public class App
{
    public static void main(String[] args) throws InterruptedException
    {
        WebDriver driver = new ChromeDriver();

        driver.manage().window().maximize();
        driver.manage().timeouts().implicitlyWait(java.time.Duration.ofSeconds(10));

        driver.get("https://automationexercise.com/");

        // Scroll down a bit
        Thread.sleep(2000);
        ((org.openqa.selenium.JavascriptExecutor) driver)
                .executeScript("window.scrollBy(0,500)");

        // Locate first product
```

```
        WebElement product = driver.findElement(By.xpath("(//div[@class='product-image-wrapper'])
[1]"));

        // Hover on product
        Actions actions = new Actions(driver);
        actions.moveToElement(product).perform();

        // Click Add to cart
        driver.findElement(By.xpath("(//a[contains(text(),'Add to cart')])[1]")).click();

        Thread.sleep(2000);

        // Click View Cart
        driver.findElement(By.xpath("//u[text()='View Cart']")).click();

        Thread.sleep(3000);

        driver.quit();
    }
}
```

2.App.java code for Practicetest Automation

```
package com.example;

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;

public class App
{
    public static void main(String[] args) throws InterruptedException
    {
        WebDriver driver = new ChromeDriver();

        // Open website
        driver.get("https://practicetestautomation.com/practice-test-login/");
        driver.manage().window().maximize();

        Thread.sleep(2000);

        // Enter username
        driver.findElement(By.id("username")).sendKeys("student");

        // Enter password
        driver.findElement(By.id("password")).sendKeys("Password123");

        // Click login
```

```
        driver.findElement(By.id("submit")).click();

        Thread.sleep(3000);

        // Click logout after login
        driver.findElement(By.linkText("Log out")).click();

        Thread.sleep(2000);

        driver.quit();
    }
}
```

3.App.java code for Saucedemo

```
package com.example;


import org.openqa.selenium.By;

import org.openqa.selenium.WebDriver;

import org.openqa.selenium.chrome.ChromeDriver;



public class App

{

    public static void main( String[] args )

    {

        WebDriver driver=new ChromeDriver();

        driver.get("https://www.saucedemo.com/");

        driver.manage().window().maximize();

        driver.findElement(By.id("user-name")).sendKeys("standard_user");

        driver.findElement(By.id("password")).sendKeys("secret_sauce");

        driver.findElement(By.id("login-button")).click();

    }

}
```

Step 5: Build the Maven Project

Run the following command:

mvn clean install

This command compiles the project and builds the JAR file.

```
brijesh@Ubuntu:~/Devops/MyMavenSeleniumApp02$ mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< com.example:MyMavenSeleniumApp02 >------------------
[INFO] Building MyMavenSeleniumApp02 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MyMavenSeleniumApp02 ---
[INFO] Deleting /home/brijesh/Devops/MyMavenSeleniumApp02/target
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenSeleniumApp02 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/brijesh/Devops/MyMavenSeleniumApp02/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ MyMavenSeleniumApp02 ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/brijesh/Devops/MyMavenSeleniumApp02/target/classes
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ MyMavenSeleniumApp02 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/brijesh/Devops/MyMavenSeleniumApp02/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:testCompile (default-testCompile) @ MyMavenSeleniumApp02 ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/brijesh/Devops/MyMavenSeleniumApp02/target/test-classes
[INFO]
[INFO] --- maven-surefire-plugin:2.22.2:test (default-test) @ MyMavenSeleniumApp02 ---
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.011 s - in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- maven-jar-plugin:3.0.1:jar (default-jar) @ MyMavenSeleniumApp02 ---
[INFO] Building jar: /home/brijesh/Devops/MyMavenSeleniumApp02/target/MyMavenSeleniumApp02-1.0-SNAPSHOT.jar
[INFO]
```

Similary perform mvn clean install for MySeleniumApp01&03 and execute using the command mvn exec:java -Dexec.mainClass="com.example.App"

Step 6: Run the Application

Run the application using:

mvn exec:java -Dexec.mainClass="com.example.App"

```
[INFO] ------------------------------------------------------------------------
brijesh@Ubuntu:~/Devops/MyMavenSeleniumApp01$ mvn exec:java -Dexec.mainClass="com.example.App"
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< com.example:MyMavenSeleniumApp01 >------------------
[INFO] Building MyMavenSeleniumApp01 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- exec-maven-plugin:3.6.3:java (default-cli) @ MyMavenSeleniumApp01 ---
```

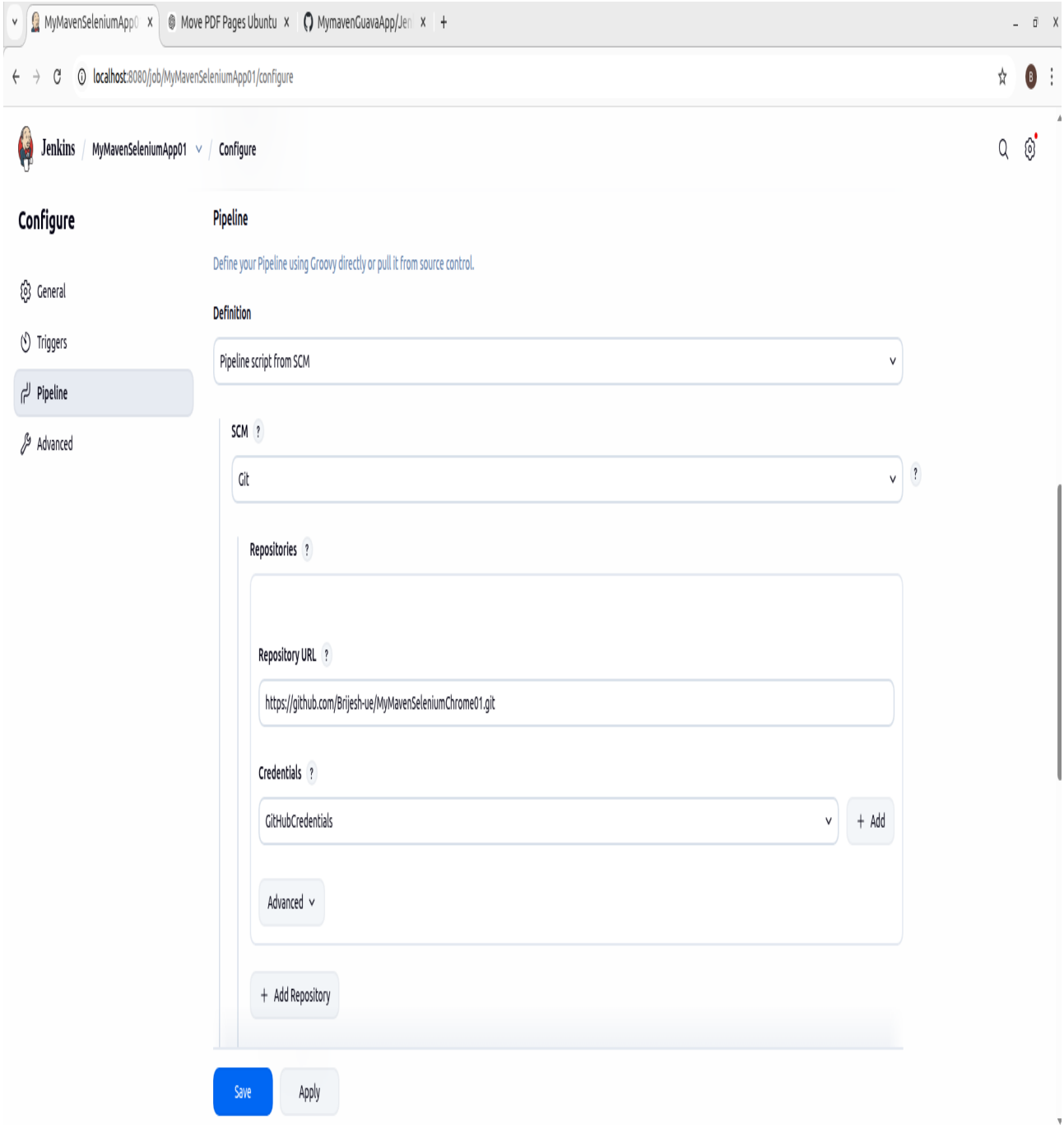
```
[INFO] ------------------------------------------------------------------------
brijesh@Ubuntu:~/Devops/MyMavenSeleniumApp02$ mvn exec:java -Dexec.mainClass="com.example.App"
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< com.example:MyMavenSeleniumApp02 >------------------
[INFO] Building MyMavenSeleniumApp02 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- exec-maven-plugin:3.6.3:java (default-cli) @ MyMavenSeleniumApp02 ---
```

```
[INFO] ------------------------------------------------------------------------
brijesh@Ubuntu:~/Devops/MyMavenSeleniumApp03$ mvn exec:java -Dexec.mainClass="com.example.App"
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< com.example:MyMavenSeleniumApp03 >------------------
[INFO] Building MyMavenSeleniumApp03 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- exec-maven-plugin:3.6.3:java (default-cli) @ MyMavenSeleniumApp03 ---
```

1.MyMavenSeleniumApp01 Pipeline:

Step 1: Create a New Pipeline Job
Go to Dashboard → + New Item.
Enter name, select Pipeline, and click OK.

Step 2:Configure Pipeline from SCM
Scroll down and select Pipeline script from SCM.
SCM: Git Enter repository URL and select GitHub credentials.



Step 3: Create Jenkinsfile

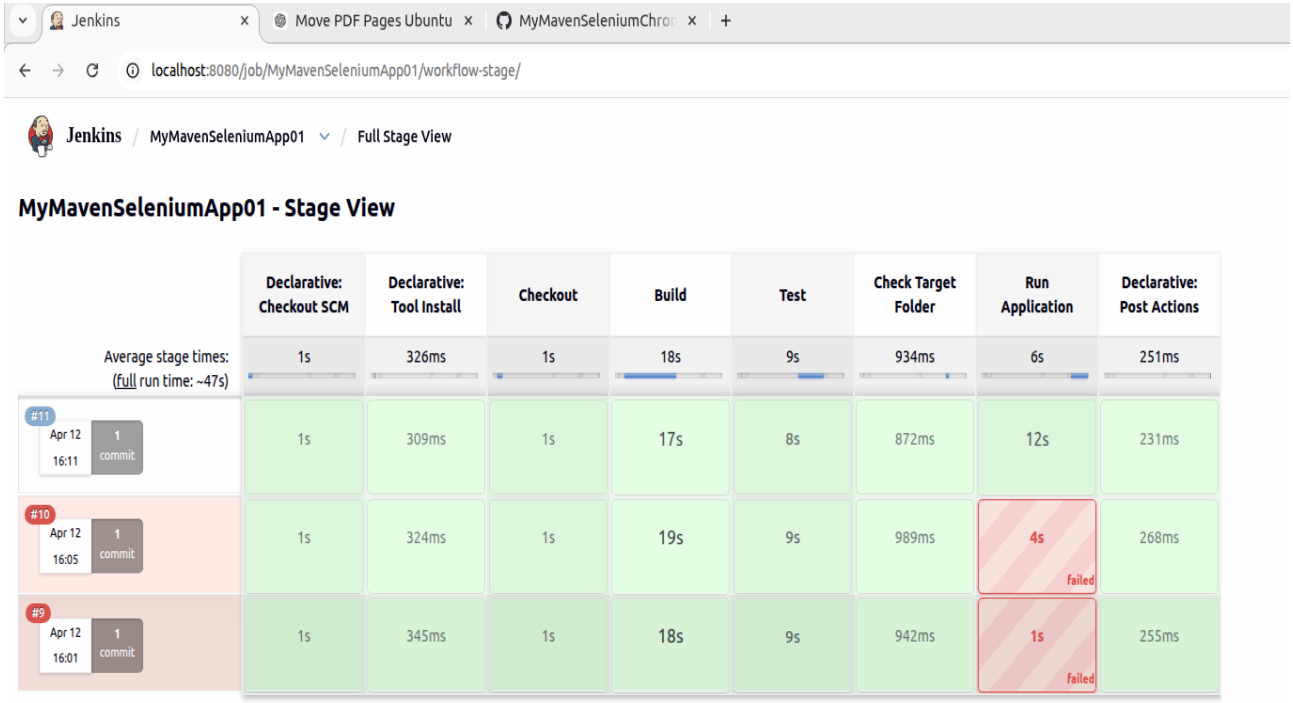
```
pipeline {
    agent any  // Use any available agent

    tools {
        maven 'Maven'  // Ensure this matches the name configured in Jenkins
    }
    stages {
        stage('Checkout') {
            steps {
                git branch: 'main', url:
'https://github.com/Brijesh-ue/MyMavenSeleniumChrome01.git'
            }
        }

        stage('Build') {
            steps {
                sh 'mvn clean package'  // Run Maven build
            }
        }

        stage('Test') {
            steps {
                sh 'mvn test'  // Run unit tests
            }
        }

stage('Check Target Folder') {
    steps {
        sh 'ls -l target'
    }
}



       stage('Run Application') {
    steps {
        sh 'java -jar target/MyMavenSeleniumApp01-1.0-SNAPSHOT.jar'
    }
```

```
}



    }

    post {
        success {
            echo 'Build and deployment successful!'
        }
        failure {
            echo 'Build failed!'
        }
    }
}
```

Step 4: Build the Pipeline

Go to Jenkins and click Build Now.

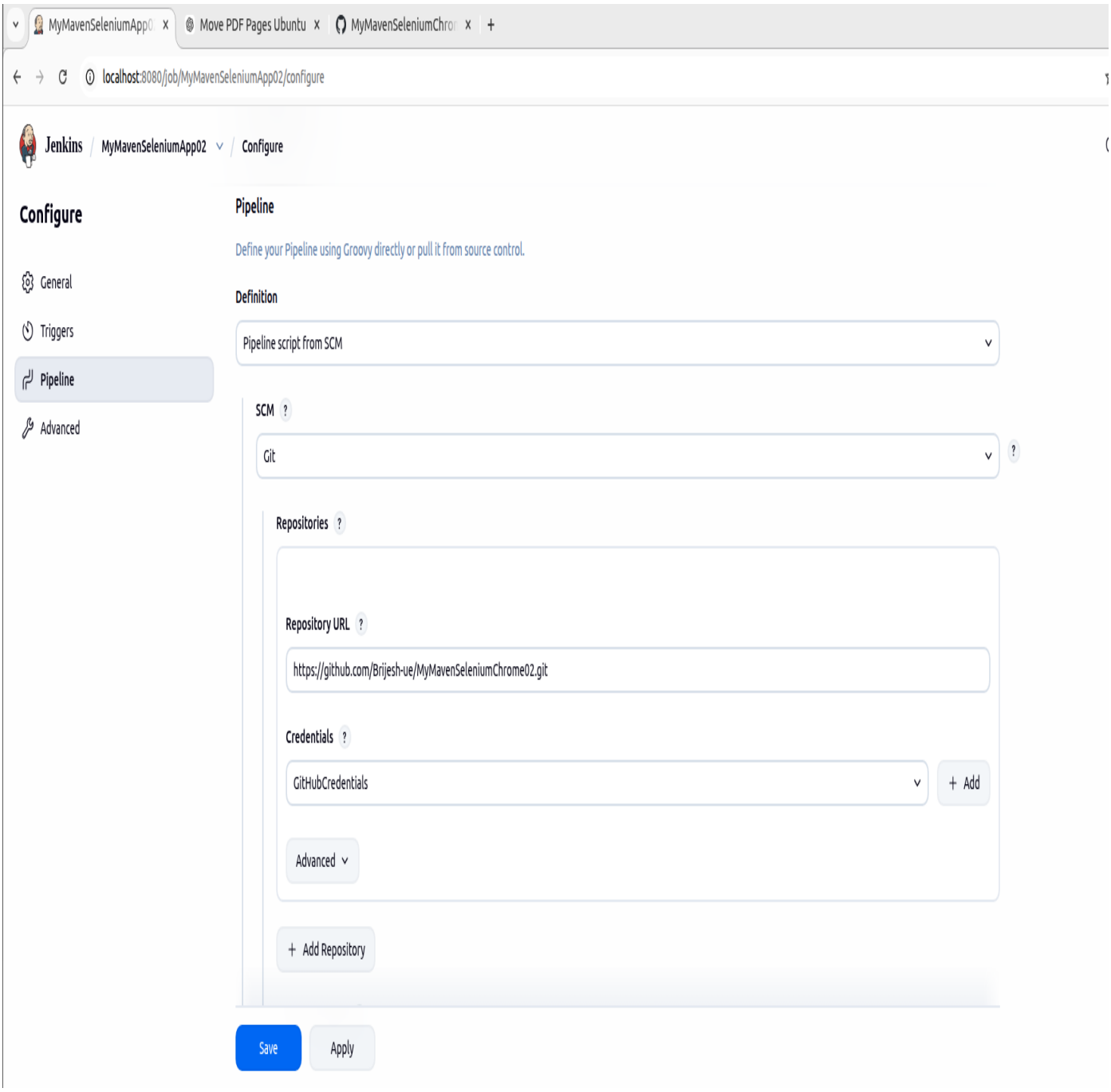
Monitor Build Stages Use Stage View and Full Stage View to monitor execution.



2.MyMavenSeleniumApp02 Pipeline:

Step 1: Create a New Pipeline Job
Go to Dashboard → + New Item.
Enter name, select Pipeline, and click OK.

Step 2:Configure Pipeline from SCM
Scroll down and select Pipeline script from SCM.
SCM: Git Enter repository URL and select GitHub credentials.



Step 3: Create Jenkinsfile

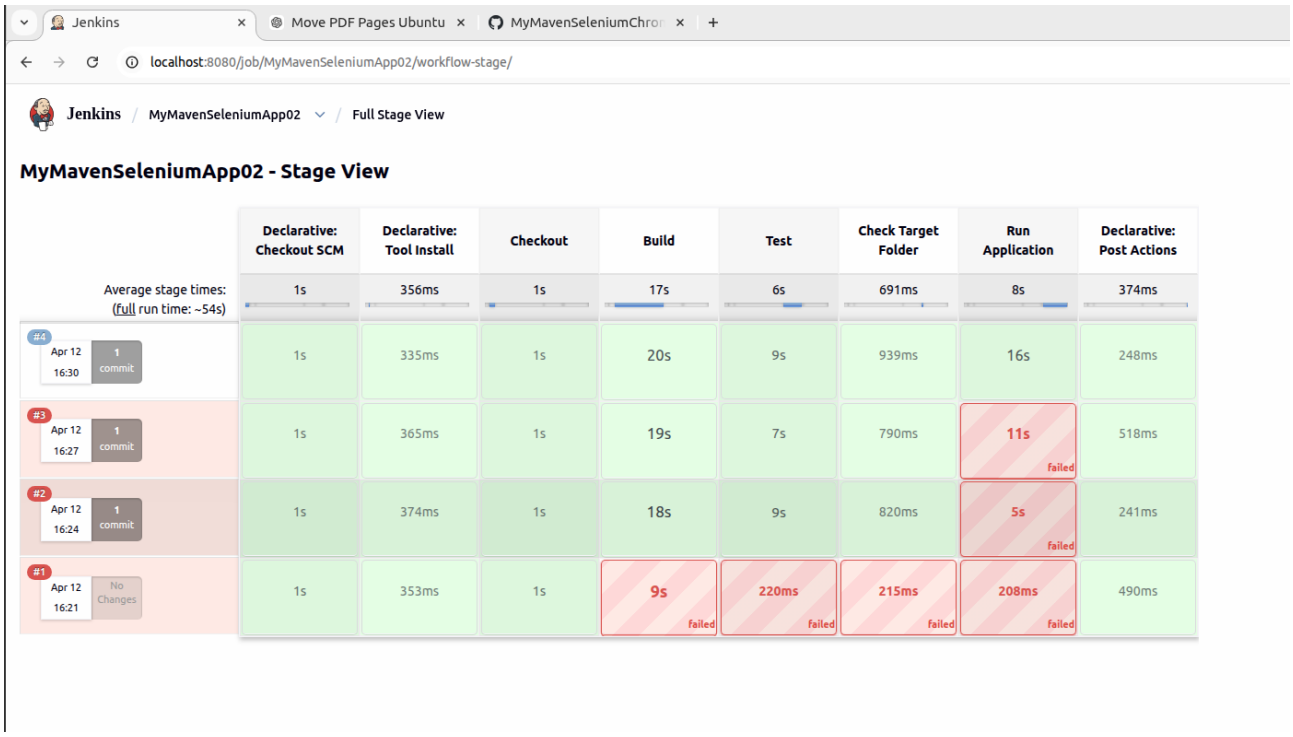
```
pipeline {
    agent any  // Use any available agent

    tools {
        maven 'Maven'  // Ensure this matches the name configured in Jenkins
    }
    stages {
        stage('Checkout') {
            steps {
                git branch: 'main', url:
'https://github.com/Brijesh-ue/MyMavenSeleniumChrome02.git'
            }
        }

        stage('Build') {
            steps {
                sh 'mvn clean package'  // Run Maven build
            }
        }

        stage('Test') {
            steps {
                sh 'mvn test'  // Run unit tests
            }
        }

stage('Check Target Folder') {
    steps {
        sh 'ls -l target'
    }
}



       stage('Run Application') {
    steps {
```

```
        sh 'java -jar target/MyMavenSeleniumApp02-1.0-SNAPSHOT.jar'
    }
}




    }

    post {
        success {
            echo 'Build and deployment successful!'
        }
        failure {
            echo 'Build failed!'
        }
    }
}
```

Step 4: Build the Pipeline

Go to Jenkins and click Build Now.

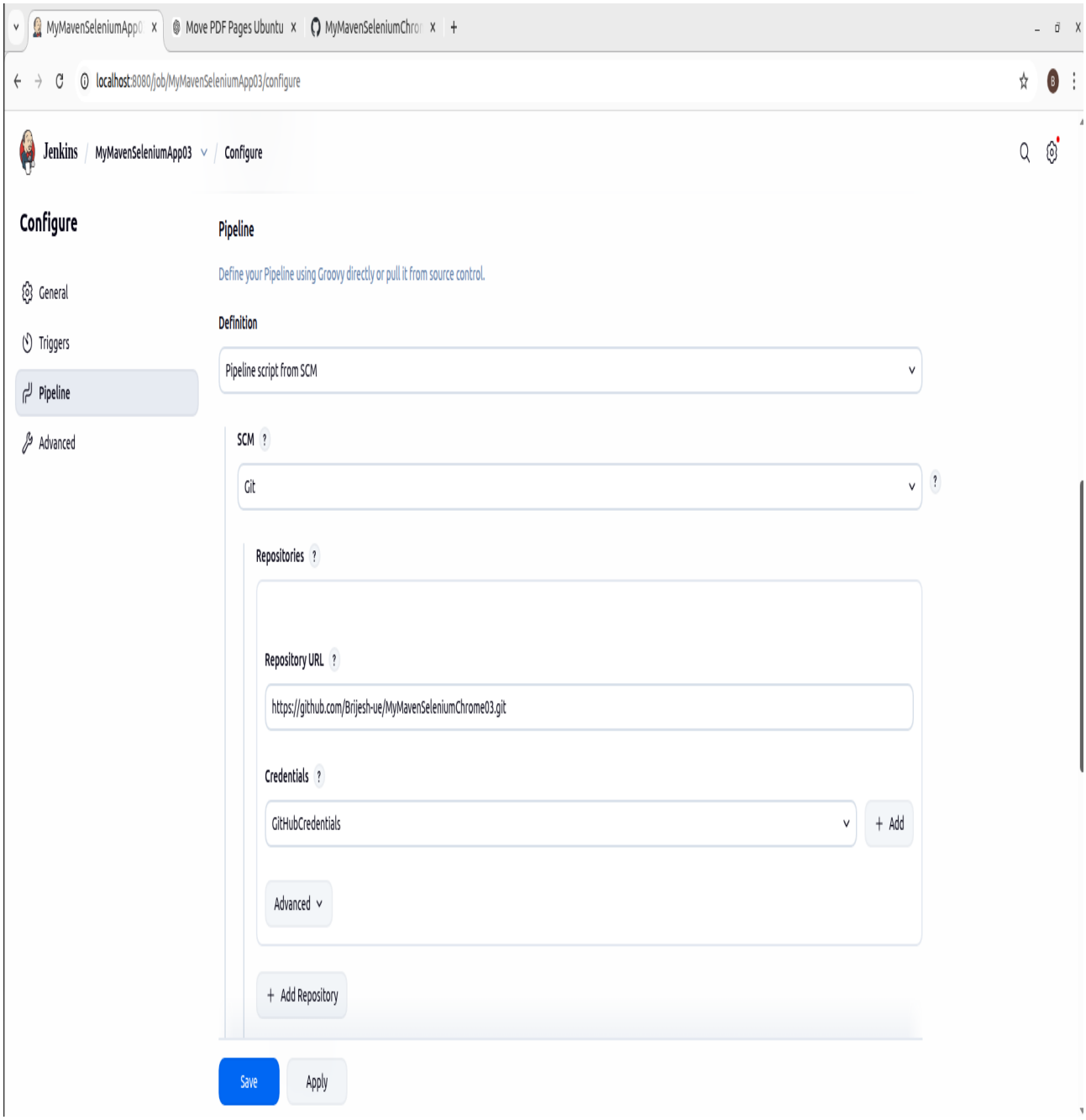
Monitor Build Stages Use Stage View and Full Stage View to monitor execution



3.MyMavenSeleniumApp03 Pipeline:

Step 1: Create a New Pipeline Job
Go to Dashboard → + New Item.
Enter name, select Pipeline, and click OK.

Step 2:Configure Pipeline from SCM
Scroll down and select Pipeline script from SCM.
SCM: Git Enter repository URL and select GitHub credentials.

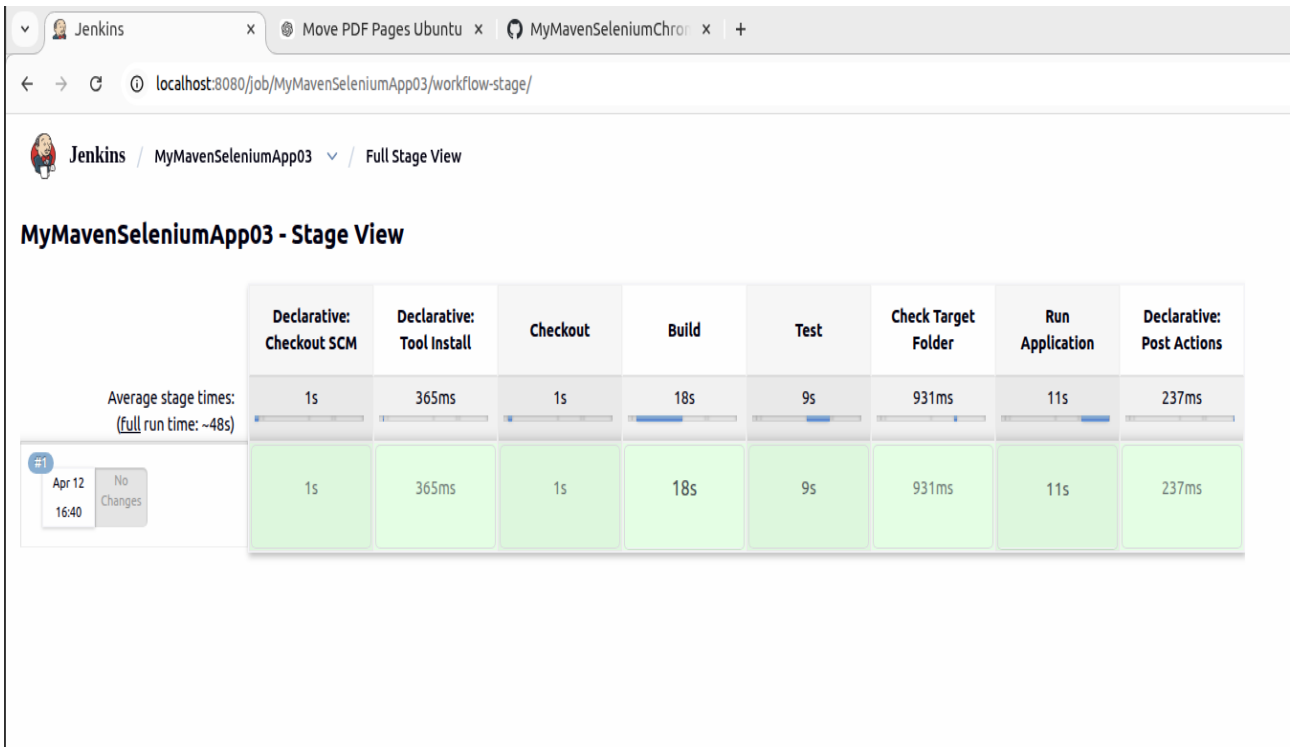


Step 3: Create Jenkinsfile

```
pipeline {
    agent any  // Use any available agent

    tools {
        maven 'Maven'  // Ensure this matches the name configured in Jenkins
    }
    stages {
        stage('Checkout') {
            steps {
                git branch: 'main', url:
'https://github.com/Brijesh-ue/MyMavenSeleniumChrome03.git'
            }
        }

        stage('Build') {
            steps {
                sh 'mvn clean package'  // Run Maven build
            }
        }

        stage('Test') {
            steps {
                sh 'mvn test'  // Run unit tests
            }
        }

stage('Check Target Folder') {
    steps {
        sh 'ls -l target'
    }
}



       stage('Run Application') {
    steps {
        sh 'java -jar target/MyMavenSeleniumApp03-1.0-SNAPSHOT.jar'
    }
```

```
}



    }

    post {
        success {
            echo 'Build and deployment successful!'
        }
        failure {
            echo 'Build failed!'
        }
    }
}
```

Step 4: Build the Pipeline

Go to Jenkins and click Build Now.

Monitor Build Stages Use Stage View and Full Stage View to monitor execution.